

Baby Words Tracker - Sprint 1 Retrospective

What Went Well

- Using firebase, firestore, and flutter made it relatively easy to get started and simplified most of the sprint 1 goals, specifically database design and authentication.
- Designing and implementing features with flutter is very easy. Flutter provides a very useful debug system with hot reloads to test changes. This simplified testing significantly.
- Integrating UI with backends was very easy due to having great backend teammates.
- Creating a user-friendly UI design with the features that our sponsor requested was fairly simple with flutter.

What Didn't Go Well

- There was some confusion with what best practice for setting up CRUD functionality with the database should look like, as many tutorials and online resources had conflicting information.
- Designing the database access functions has been a continued pain point for our team. It is difficult to find best practices here, and time consuming to correctly implement functions.
- High level of abstraction led to a few instances of hard debugging.
- Ran into some permission errors accessing the database with our current firestore security rules.

What Could Be Improved

- The current database design could lead to several security issues as we work to make the authentication process more secure, so it will be beneficial to make some minor changes (like adding parent email to each child) throughout the next sprint.
- The current database structure does not handle errors or null returns very well. It needs to be updated to be more robust to failure.
- The stats page is still in a very MVP state, it could have a better UI and way more features, and it will soon.
- The UI can be improved to be more personalized for each user and more consistent across the pages. Adding more features that parents would be interested in such as a word bank of known words, overall word count, and words by parts of speech will improve the user experience of the app.

Challenges

- Most dictionary APIs are expensive (~\$100/month), so finding a reasonably priced/free dictionary API came with some difficulty. We were eventually able to find a free API, but

it does not provide support for other languages which will become an issue when we add spanish features.

- Synchronizing authentication with the database is difficult and requires an admin service app. It will also be necessary to add custom claims before authentication can be considered secure.
- For our audio processing stretch goal, finding an API that can accurately make out baby words is just that – a stretch.
- Understanding how change notifiers work and implementing database access functions has been challenging.